

Resolve A with respect to new polarization axis. It is $A \cos 60^\circ = 0.50 A$

16 C

Phase difference $\phi = \frac{t}{T} \times 2\pi = \frac{4}{24} \times 2\pi = \pi/3$

For time-axis, whichever (crest) is behind is leading.

17 B

As α particle moves near nucleus, it experiences an electric force of repulsion that